

Sara Chaudhari

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EDUCATION

University of California, San Diego , MS in Computer Science and Engineering	Sept 2024 – Jun 2026
• Relevant Coursework: Distributed Systems, Machine Learning Systems, Scalable Data Systems, Advanced Algorithms, Recommendation Systems & Web Mining, Statistical NLP, Generative AI	GPA: 3.9/4.0
University of Mumbai , BE in Computer Engineering	Aug 2020 – Jun 2024
• Honors in Artificial Intelligence and Machine Learning	GPA: 4.0/4.0

EXPERIENCE

Software Developer Intern, Genies	Sept 2025 – Dec 2025
• Developed distributed semantic retrieval services in Java for large-scale conversational AI systems, implementing embedding ranking and metadata indexing pipelines that improved routing accuracy 37%.	
• Built scalable backend inference services using concurrency primitives and caching strategies, reducing request latency 15% while improving reliability of high-traffic conversational AI systems.	
• Designed monitoring, validation, and CI/CD deployment pipelines, improving debugging visibility and reducing production deployment failures 25% across distributed backend infrastructure supporting conversational AI services.	

Software Engineer Intern, Dassault Systèmes Americas Corp.	June 2025 – Sept 2025
• Designed and developed scalable backend APIs and workflow services using microservice architecture, processing 2,000+ product lifecycle operations daily across enterprise engineering collaboration platforms.	
• Built containerized rule-processing pipelines using Python, Docker, and Kubernetes, enabling distributed evaluation of engineering workflow data and reducing pipeline runtime 22% across large-scale datasets.	
• Implemented automated regression testing infrastructure using Groovy with Jenkins CI/CD pipelines, increasing QA coverage and preventing production regressions across distributed enterprise backend services.	

Software Engineer, UpSolve Solutions	Apr 2023 – Jul 2024
• Built ML inference APIs using Python and AWS infrastructure, serving 5,000+ real-time prediction requests daily supporting recommendation systems and analytics workloads in production environments.	
• Designed asynchronous data ingestion pipelines using Redis queues and AWS SQS, improving system throughput and ensuring reliable processing of high-volume machine learning workloads during peak traffic.	
• Optimized database indexing strategies and query execution in PostgreSQL, improving code performance and reducing feature lookup latency 30% across production analytics and recommendation systems.	

Software Engineer, Garg Group	Sept 2022 – Apr 2023
• Developed anomaly detection pipelines for supply-chain time-series data using LSTM neural networks, improving anomaly detection accuracy 15% across warehouse inventory monitoring systems.	
• Built backend ingestion services using FastAPI and AWS Lambda, processing warehouse telemetry streams and enabling reliable data collection for downstream machine learning analytics systems.	
• Designed feature engineering workflows for logistics datasets, improving training data quality and enabling more reliable anomaly detection across warehouse inventory monitoring pipelines.	

RESEARCH WORK

Graduate Researcher, UC San Diego	Mar 2025 – Sept 2025
• Developed an experimental framework for evaluating machine learning and reinforcement learning training pipelines, implementing modular data ingestion workflows supporting 40+ experiments across datasets exceeding 2 million records.	
• Designed reproducible distributed training environments using Docker and Kubernetes, enabling controlled experimentation across 20+ training configurations and reducing environment setup time 35%.	

KEY PROJECTS

Distributed Log Indexing and Search System
• Designed and implemented a distributed log indexing system using Java multithreading and shard-based storage, indexing 50GB+ application logs while supporting concurrent query execution across distributed nodes.
• Implemented parallel query processing and optimized inverted index structures, achieving 3× faster indexing throughput and improving log retrieval accuracy for developer debugging tools.

Hybrid Semantic Retrieval and Ranking System for Large Document Collections
• Developed hybrid search architecture combining BM25 lexical retrieval and FAISS vector indexing, enabling scalable semantic search across 1 million documents for information retrieval evaluation workloads.
• Evaluated ranking performance using NDCG@10, MAP, and Recall@50, achieving 32% improvement in NDCG@10 and 18% higher recall compared with baseline lexical retrieval models.

PUBLICATIONS

- **Genome Sequence Analysis and Drug – Target Interaction Prediction using Deep Learning** (Link)
Journal: Springer Innovative Computing and Communications, LNNS Series (Volume 1038)
- **Two-Pass Pipeline for Automated Medical Infographic Generation Using Text Summarization**
Presented in SmartDataCom: International Conference on Smart Data Processing, Communication and Networking (2024)

SKILLS

Programming: Java, C++, Go, Python, SQL, Bash
Backend & Systems: Distributed Systems, Multithreading, Concurrency, RESTful API Design, Microservices Architecture, System Design, Linux
Data & Databases: PostgreSQL, MongoDB, Data Modeling, Query Optimization, Indexing, Data Pipelines
Machine Learning Systems: PyTorch, TensorFlow, Information Retrieval, Ranking Systems, Embeddings, Natural Language Processing
Infrastructure & Tools: Docker, Kubernetes, AWS, Google Cloud Platform (GCP), Jenkins, Git, CI/CD Pipelines